



BANGLADESH TECHNICAL EDUCATION BOARD
Agargaon, Sher-E-Bangla Nagar
Dhaka-1207.

04-YEAR DIPLOMA IN ENGINEERING CURRICULUM
COURSE STRUCTURE & SYLLABUS
(PROBIDHAN-2022)

COMPUTE SCIENCE AND TECHNOLOGY
TECHNOLOGY CODE: 85

7th SEMESTER
(Effective from 2022-2023 Academic Sessions)

Subject Code	Subject Name	Period/Week		Credit
65853	INNOVATION AND ENTREPRENEURSHIP	T	P	C
		2	0	2

Rationale	Innovation is production or adoption, assimilation, and exploitation of a value-added novelty in economic and social spheres; renewal and enlargement of products, services, and markets; development of new methods of production; and the establishment of new management systems. It is both a process and an outcome. On the contrary, an entrepreneur is an individual who creates a new business, bearing most of the risks and enjoying most of the rewards. The process of setting up a business is known as entrepreneurship. The entrepreneur is commonly seen as an innovator, a source of new ideas, goods, services, and business or procedures. Innovation and Entrepreneurship is part and parcel in engineering arena. This is because without innovative knowledge, skill and attitude an entrepreneur or an engineer is merely an entrepreneur or simply an engineer. But having sound knowledge on innovation an engineer or entrepreneur may be a successful person or figure in the respective field. Besides innovation is more appropriate to involve the practical implementation of an invention (i.e.- new or improved ability) to make a meaningful impact in a market or society and not all innovations require a new invention. Technical innovation often manifests itself via the engineering process when the problem being solved is of a technical or scientific nature. Due to the high risks involved in launching a start-up, a significant proportion of start-up businesses have to close due to "lack of funding, bad business decisions, government policies, an economic crisis, lack of market demand, or a combination of all of these." So, Syllabus is designed to cover up the challenges and steps to become an entrepreneur. That's why stress is given how to make a project profile and how to prepare a business plan. In addition, to overcome other associated problems, everything which is essential are tried to incorporate accordingly.
Learning Outcomes (Theoretical)	After undergoing the subject, students will be able to ➤ Describe Innovation and right mindset for Innovation. ➤ State Innovation steps. ➤ Explain types of Innovation and product stages. ➤ Describe the areas of service Innovation. ➤ State the prospect of Entrepreneurship development in Bangladesh. ➤ Discuss the idea of project. ➤ Explain the SWOT analysis. ➤ Describe technical aspects of a project. ➤ Discuss short-term and long-term financial plan. ➤ Explain Business plan and its contents. ➤ Describe Venture ideas and sources of assistance. ➤ Explain selection of venture idea and sanctioning procedure for assistance. ➤ State Insurance and Premium. ➤ Describe self- employment, MDG, SDG, Green TVET culture.

Detailed Syllabus (Theory)

Unit	Topics with Contents	Period	Marks
1.	INNOVATION 1.1 Define Innovation. 1.2 Explain the Aims of Innovation. 1.3 Explain the importance of Innovation. 1.4 Explain TCV (Time, Cost, Visit) in Innovation. 1.5 Describe developing the innovative Mindset.	2	5
2	INNOVATION PROCESS 2.1 State the steps of Innovation. 2.2 Describe the Idea Generation & Mobilization of Innovation. 2.3 Mention the Advocacy & Screening of Innovation. 2.4 Explain the Experimentation of Innovation. 2.5 Describe the Commercialization of Innovation. 2.6 Explain Diffusion & Implementation of Innovation.	3	7
3	TYPES OF INNOVATION 3.1 Classify Innovation. 3.2 Define Radical Innovation. 3.3 Describe the Incremental Innovation. 3.4 Mention the types of Product Innovation. 3.5 Explain the Product Innovation stages. 3.6 Define the service Innovation. 3.7 Describe the area of Service Innovation. 3.8 Explain the Process Innovation.	3	8
4	ENTREPRENEURSHIP AND ECONOMIC DEVELOPMENT 4.1 Define entrepreneurship and entrepreneur. 4.2 Discuss the characteristics and qualities of an entrepreneur. 4.3 Mention the classification of entrepreneur. 4.4 Discuss the necessity of entrepreneurship. 4.5 Discuss the prospect of entrepreneurship development in Bangladesh. 4.6 Define economic development and economic growth. 4.7 Discuss the economic development process. 4.8 Discuss the capital accumulation or rate of savings.	4	7
5	PROJECT SELECTION PROCESS & FINANCIAL PLANNING 5.1 Define project. 5.2 Discuss the idea of a project. 5.3 Explain SWOT analysis. 5.4 Describe the guide lines for project ideas. 5.5 Discuss the sources of project ideas. 5.6 Discuss the evaluation of project ideas. 5.7 Describe the technical aspect of a project. 5.8 Define financial planning. 5.9 Discuss the long-term financial plan. 5.10 Discuss the short-term financial plan.	4	8
6	BUSINESS PLAN AND ENVIRONMENT FOR ENTREPRENEURSHIP.	4	6

	6.1 Define business plan. 6.2 State the different steps to prepare a Business Plan. 6.3 Discuss the contents of business plan. 6.4 State the necessary business documents for Business Plan. 6.4 Define environment of business. 6.5 Describe the influencing factors of environment on entrepreneurship. 6.6 Discuss political, socio-cultural, economical, legal and technological environment. 6.7 Explain micro and macro environment.		
7	VENTURE IDEAS, ASSISTANCE, SANCTIONING PROCEDURE AND EVALUATION 7.1 Describe sources of venture ideas in Bangladesh. 7.2 Discuss different types and sources of venture ideas in Bangladesh. 7.3 Define sources of assistance. 7.4 Describe different types of sources of assistance. 7.5 Discuss the aid of sources. 7.6 Discuss the industrial policy. 7.7 Define foreign aid. 7.8 State evaluation of venture ideas. 7.9 Discuss the considering factors in the selection of venture idea.	4	5
8	INSURANCE AND PREMIUM 8.1 Define insurance and premium. 8.2 Describe the essential conditions of insurance contract. 8.3 Discuss various types of insurance. 8.4 Distinguish between life insurance and general insurance.	2	5
9	SELF EMPLOYMENT 9.1 Define self-employment. 9.2 Describe different types of employment. 9.3 Describe the importance of business as a profession. 9.4 Discuss the reasons for success and failure in business.	2	4
10	SUSTAINABLE DEVELOPMENT GOALS (SDG) 10.1 Define MDG and SDG. 10.3 Mention the objectives of SDG. 10.2 List the 17 goals of SDG. 10.4 Explain the actions to face the challenges of SDGs. 10.5 Mention the targets of SDG 4 and SDG 8. 10.6 State the elements of Quality education for TVET. 10.7 Describe the gender equity and equal access of TVET for economic growth. 10.8 Explain Green development, Green Economy, Green TVET and Green jobs. 10.9 Explain the role an entrepreneur for achieving SDG.	4	5
	Total	32	60

Recommended Books:

Sl	Book Name	Writer Name	Publisher Name & Edition
01	Innovation and Entrepreneurship	Peter F Drucker	Routledge Classic
02.	Innovation and Entrepreneurship	Otto Chui Chau Lin	World Scientific https://doi.org/10.1142/10195 April 2018
03.	Entrepreneurship and Innovation	Tim Mazzarol	Tilde University Press
04.	Entrepreneurship: The Practice and Mindset	Heidi M. Neck Christopher P. Neck Emma L. Murray	Sage Pubns
05.	The Art of Innovation	Tom Kelley, Jonathan Littman Tom Peters	New York Currency/Doubleday 2001
06	Creativity, Innovation, and Entrepreneurship	H. James Harrington	
07	Creativity and Innovation in Entrepreneurship	S.S Khanka	Sultan chand sons, 2021
08	Entrepreneurship Business and Management	Dr. R.C Bhatia	Sultan chand sons, 2020

Name and particulars of the participants:

SL. NO	NAME AND DESIGNATION AND MOBILE NO	SIGNATURE
1	Engr. Pradipta Khisa, Principal (In-charge), Feni Polytechnic Institute, Feni. Mob: 01715-747546	
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4	S. M. Shahjahan, Deputy Director (Course Accreditation), BTEB, Dhaka. Mob: 01712-125765	

Subject Code	Subject Name	Period per Week		Credit
28571	Digital Marketing Technique	T	P	C
		2	3	3

Rationale	The objective of this subject Digital Marketing Techniques course is to equip students with a comprehensive understanding of the key components and strategies essential for successful digital marketing.
Learning Outcome (Theoretical)	After undergoing the subject, students will be able to: 1. Explain the procedure to create Facebook page and YouTube channel. 2. Interpret technique of effective digital marketing campaigns. 3. State the process to student online engagement. 4. Explain the creation of branding. 5. Elucidate the mechanism to measure the outcome of online activities.
Learning Outcome (Practical)	After undergoing the subject, students will be able to: 1. Create Text Content Using AI 2. Create an Image Using Canva 3. Create a simple Video for Digital Marketing 4. Demonstrate Keyword Research 5. Create Profile Backlinks 6. Create Facebook Page 7. Create a YouTube Channel 8. Create and optimize a LinkedIn Profile 9. Create a Google Ads Account 10. Perform Email Marketing Campaign

Detailed Syllabus (Theory)

Unit	Topics with Contents	Class (1 Period)	Final Marks
1	Introduction of Digital Marketing 1.1. Define Digital Marketing 1.2. Mention the Importance of Digital Marketing. 1.3. Differentiate between Digital Marketing and Traditional Marketing. 1.4. Describe Current Trends and Statistics in Digital Marketing. 1.5. State Key Components of Digital Marketing. 1.6. Describe the Careers in Digital Marketing.	1	3
2	Content for Digital Marketing 2.1 Mention the Importance of Content in Digital Marketing. 2.2 Mention the different types of Content. 2.3 Describe Text, Images, Videos, Infographics content. 2.4 Explain Strategies & Tools of Content Creation.	2	5
3	Search Engine Optimization (SEO) 3.1 Define SEO. 3.2 Mention the Importance of SEO in Digital Marketing. 3.3 Explain Local & E-commerce SEO. 3.4 Describe Keyword Research and Optimization. 3.5 Explain On-page SEO Techniques. 3.6 Explain Off-page SEO Techniques. 3.7 Explain Technical SEO Techniques. 3.8 Describe the uses of Google Search Console tools for website Performance 3.9 State the Uses Of GA4 For Website Tracking	10	15
4	Social Media Marketing (SMM) 4.1 Define Social Media Marketing. 4.2 Mention the Importance of SMM in Digital Marketing. 4.3 Explain Social Media Content Creation.	1	3
5	Facebook Marketing Strategies 5.1 State the Procedure to Create and Optimization of Facebook Page. 5.2 Describe the Procedure to Create and Optimization Facebook Group. 5.3 Explain Procedure to Create and Maintenance Facebook Ads and Campaigns.	5	10
6	YouTube Marketing Strategies 6.1 State the Procedure to Create and Optimization Of YouTube Channel. 6.2 Describe the process to Upload Video According to YouTube Guideline 6.3 Explain the technique to Optimize Video for Better View and User Experience. 6.4 Explain the procedure to Video Marketing on Youtube.	5	5
7	LinkedIn Marketing Strategies 7.1 State the Procedure to Create and Optimization of LinkedIn Profile. 7.2 State the Procedure to Create and Optimization of LinkedIn Company Page. 7.3 Explain Professional Networking on LinkedIn.	3	5

8	Search Engine Marketing (SEM) 8.1 Define SEM. 8.2 Differentiate between SEO and SEM. 8.3 Describe Google Ads Overview. 8.4 Explain Types of Google Ads.	2	4
9	Email & Tele Marketing 9.1 Define Email Marketing. 9.2 Mention the Importance of Email Marketing in Digital Marketing 9.3 Describe Email Marketing Tools and Platforms. 9.4 State Telemarketing. 9.5 Mention the Importance of Telemarketing in Digital Marketing. 9.6 Describe the Telemarketing Strategies.	2	7
10	Ethics in Digital Marketing 10.1 Define Ethics of Digital Marketing. 10.2 State the Importance of Ethics in Digital Marketing. 10.3 Mention Common Ethical Issues in Digital Marketing. 10.4 Describe Ethical Guidelines for Creating Content. 10.5 State the ICT policy in Bangladesh. 10.6 Mention International Organization of Ethics of Digital Marketing.	1	3
	Total	32	60

DETAILED SYLLABUS (PRACTICAL)

SL.	EXPERIMENT NAME WITH WORKING PROCEDURE	Class (3 Period)	Marks (Continuous)
1	Create Text Content Using AI 1. Choose a specific topic for the content. 2. Access the AI interface on OpenAI's website or platform. 3. Enter a detailed prompt specifying the topic and desired length of the text content. 4. Read through the AI-generated text for accuracy and relevance. 5. Make necessary edits to improve clarity, grammar, and coherence. 6. Save the final text content in a word processor or content management system.	1	2
2	Create an Image Using Canva or any other tools 1. Determine the purpose and size of the image (e.g., blog header, social media post). 2. Open Canva and log in or sign up for an account. 3. Choose a template that suits your needs or start with a blank canvas. 4. Customize design 5. Check the overall design for consistency and balance. 6. Download the image in the desired format (e.g., PNG, JPG).	1	2
3	Create a simple Video for Digital Marketing 1. Pick a video topic relevant to your brand and target audience 2. Make it short and engaging, ideally under 1 minute. 3. Collect smartphone camera, video recording app, props, visuals, background elements.	2	3

	4. Prepare video with ensuring good lighting and clear audio, shoot in landscape mode. 5. Edit your video using free editing tools to trim clips, add music/text. 6. Optimize your video with a clear title and sub-title.		
4	Demonstrate Keyword Research 1. Determine the primary goal for the keyword research 2. Identify blog content, product page and services. 3. Identify initial seed keywords relevant the topic or business based on the Brainstorm. 4. Apply Keyword Research Tools. 5. Analyze keywords and its competition. 6. Create a keyword research document or spreadsheet to organize your target keywords.	2	2
5	Create Profile Backlinks 1. Analyze high-authority websites, directories, and forums related on profile creation 2. Collect necessary information regarding business or personal brand. 3. Register for accounts on the identified websites. 4. Complete profiles 5. Optimize profiles for better visibility and engagement	2	2
6	Create Facebook Page 1. Visit the Facebook URL: www.facebook.com 2. Create a new account. 3. Sign in to Facebook. 4. Navigate to the "Pages" section and click "Create New Page" 5. Enter the company's name, category, and a brief description 6. Upload a profile picture (e.g., company logo) and a cover photo that represents the brand 7. Add business details such as contact information, address, and website URL 8. Create a username for the page (e.g., @CompanyName) 9. Add a call-to-action (CTA) button (e.g., "Contact Us," "Shop Now") 10. Post a welcome message and pin it to the top of the page 11. Invite friends and customers to like and follow the page	2	2
7	Create a YouTube Channel 1. Sign in to YouTube with your Google account. 2. Create a new channel with a clear, brand-focused name. 3. Upload a high-quality logo for your profile picture. 4. Design a visually appealing banner image for your channel. 5. Write a compelling description summarizing your content (max 1000 characters). 6. Include relevant links to your website or social media profiles. 7. Optimize video uploads with titles, descriptions, thumbnails, and tags. 8. Customize your channel layout, create playlists, and explore sections. 9. Promote your channel on social media, your website, or through ads.	1	2
8	Create and optimize a LinkedIn Profile 1. Sign in to LinkedIn with your existing account or create a new one.	2	4

	- Click "Me" and select "View profile" or "Create profile." - Enter your current job title, company, and location. - Upload a professional headshot for your profile picture. - Craft a compelling headline that summarizes your expertise. - Write a detailed summary highlighting your skills and experience. - Add your past work experiences, including company names, job titles, and dates. - List your educational background with institutions, degrees, and years of attendance. - Include relevant skills and expertise, using keywords from your industry. - Add volunteer work, certifications, or projects to showcase your well-roundedness.		
9	Create a Google Ads account - Sign in to your Google account or create a new one. - Visit Google Ads platform and click "Start Now." - Enter your business name, website URL (if available), and a brief description. - Select your advertising goals (e.g., website traffic, leads, sales). - Set your daily or campaign budget. - Choose your target audience demographics and interests. - Craft compelling Ads copy and select visuals. - Review your campaign details and confirm payment information.	1	2
10	Conduct Email Marketing Campaign - Set your target for email marketing. - Choose an email marketing platform based on needs. - Grow your email list by encouraging website visitors, social media followers, or existing customers to subscribe. - Group subscribers based on interests or demographics to personalize emails. - Write clear, concise, and valuable emails that resonate with your audience. - Design visually appealing emails using templates, images, and CTAs (calls to action). - Schedule and send campaigns at optimal times. - Track email open rates, click-through rates, and conversions to measure success.	2	4
	Total	16	25

Recommended Books:

SL	Book Name	Writer Name	Publisher Name & Edition
1	Digital Marketing Essentials	by Larson and Draper	Stukent, Incorporated
2	Advanced Digital Marketing Strategies	by Jones and Smith	
3	Online resources: HubSpot Academy, Google Digital Garage, industry reports and case studies		

WEBSITE REFERENCES:

SL	Web Link
1	Digital Marketing Tutorial - Java point
2	Digital Marketing Tutorial (tutorialspoint.com)
3	Best Digital Marketing Courses Online with Certificates [2024] Coursera
4	What Is Digital Marketing? Types, Strategies & Best Practices – Forbes Advisor
5	Digital marketing - Search Videos (bing.com)

Subject Code	Subject Name	Period/Week		Credit
28572	Network Administration & Services	T	P	C
		3	3	4

Rationale	Computer Networking is the most significant area of Diploma in Engineering in Computer Science & Technology. To work with Computer Networking should have the knowledge, skills and attitude of Computer Network Fundamentals, Topologies, Sub-Netting, Routing and various types of Protocols. In addition these are important to acquire sufficient knowledge and skills in the fields of Automation in Networking, Virtual LAN Routing, Network Address Translator, Network Addressing and Client Server Network to provide Domain and IP services of Network.
Learning Outcome (Theoretical)	After Completing the subject, students will be able to: 1. State Computer network basics. 2. Explain Network topologies. 3. Explain Sub-netting, VLSMs, and Summarization 4. Internet Protocol Routing 5. Describe and Configure Open Shortest Path First 6. Describe Enhanced Interior Gateway Routing Protocol 7. Explain Automation in Networking 8. Describe and Configure Various IP & Domain Services 9. Illustrate Spanning Tree Protocol, 10. Explain & Configure VLANs and Inter-VLAN routing. 11. Describe Internet Protocol Version 6 12. Illustrate & Configure Link and Gateway Redundancy
Learning Outcome (Practical)	After undergoing the subject, students will be able to: 1. Perform the Design, Development and Simulation of a network according to Three Tier architecture 2. Configure remote access Services 3. Simulate of DHCP server Configuration 4. Configure various Routing Protocol 5. Perform the design & implementation of Domain Controller with group policy 6. Configure Hot Standby Router Protocol 7. Configure Network Address Translator 8. Perform Access Control List Configuration. 9. Perform the Design, Develop and Simulation of Ether-channel 10. Configure the various properties of Spanning Tree Protocol 11. Perform the Design, Develop and Simulation of Inter- VLAN Routing

Detailed Syllabus (Theory)

Unit	Topics with Contents	Class (1 Period)	Marks
1	Network Theories 1.1 Define Network Administration 1.2 Mention the Key Components of Network Administration 1.3 Illustrate the role of network administrator 1.4 Describe Intranet, Extranet, Internet 1.5 Describe TCP/IP model 1.6 Define Data encapsulation of TCP/IP model. 1.7 Illustrate Network Hierarchical model – Two Tier, Three Tier, Spine Leaf	3	10
2	Sub-Netting, VLSMs, and Summarization 2.1 Define Sub-netting 2.2 State Classless inter-domain routing (CIDR) 2.3 Define Variable length subnet mask (VLSMs) 2.4 Describe VLSM design and implementing VLSM Network 2.5 State Summarization. 2.6 State the process of Sub-netting.	4	10
3	IP Services 3.1 State the IP Services 3.2 Explain IP Routing in network. 3.3 Discuss types of routing protocols. 3.4 Describe DNS service 3.5 Illustrate DHCP service 3.6 Describe Telnet 3.7 Discuss SSH	4	10
4	Internet Routing Protocol 4.1 Define Routing. 4.2 Explain IP Routing in network. 4.3 Discuss types of routing protocols. 4.4 Describe OSPF terminology. 4.5 Define OSPF operation. 4.6 Describe Loopback interface	3	10
5	Domain Services 5.1 Explain Workgroup & Domain Environment 5.2 Describe Active Directory Domain Service 5.3 Define group policy 5.4 Mention the types of group policy 5.5 Explain the purpose of group policy	4	10
6	Network Security Fundamentals 6.1 Define key security concepts(Threats, Vulnerabilities, Exploits and Mitigation Techniques) 6.2 Describe security program elements(User awareness, Training and Physical Access Control) 6.3 Describe security password policies elements(Management, Complexity and Password alternatives) 6.4 Describe Virtual Private Network(VPN) 6.5 Mention the types of VPN 6.6 Explain each type of VPN 6.7 Compare authentication, authorization and accounting(AAA) 6.8 Describe confidentiality, integrity and availability(CIA)	5	20

7	Spanning Tree Protocol (STP) 7.1 Define Spanning Tree Protocol 7.2 Describe Spanning Tree cost calculation 7.3 Define Spanning Tree port states 7.4 State Spanning Tree port fast 7.5 Define Rapid Spanning Tree 7.6 State Spanning Tree BPDU Guard 7.7 Define Spanning Tree BPDU Filter 7.8 Define Spanning Tree Root guard 7.9 State Ether-channel	5	15
8	VLANS and Inter-VLAN routing 8.1 Define VLANs 8.2 State 802.1Q and ISL Encapsulation. 8.3 Define Trunk link 8.4 Describe Router on a Stick. 8.5 Describe Inter-VLAN routing by multilayer switch 8.6 State the VLAN Trunking protocol (VTP)	4	10
9	Network Address Translator (NAT) 9.1 State Network Address Translator (NAT) 9.2 Explain the functions of NAT 9.3 Classify Network Address Translator 9.4 Describe each type of Network Address Translator	2	10
10	Access Control List 10.1 State access control list (ACL). 10.2 Explain the functions of access control list 10.3 Classify the access control list 10.4 Explain each type of access control list	3	10
11	Internet Protocol Version 6 (IPv6) 11.1 Mention the benefits and uses of IPv6 11.2 Define IPv6 addressing and expression 11.3 Describe how IPv6 works in an Inter-network. 11.4 State IPv6 Routing protocols (RIPng, EIGRP, OSPFv3).	3	10
12	Link and Gateway Redundancy 12.1 Define Link Redundancy 12.2 Mention the types of Link Redundancy 12.3 Define Hot Standby Router Protocol (HSRP) 12.4 State Virtual Router Redundancy Protocol (VRRP) 12.5 Describe Gateway Load Balancing Protocol (GLBP)	4	15
13	Automation in Networking 13.1 Explain how automation impacts in networking management 13.2 Compare traditional network & Controller based networking 13.3 Describe Software Defined Networking 13.4 Discuss control plane and data plane 13.5 Discuss on northbound and southbound 13.6 Recognize the capabilities of configuration management mechanisms (Puppet, Chef and Ansible)	4	10
Total		48	150

Detailed Syllabus (Practical)

Sl.	Experiment Name with Procedure	Class (3 Period)	Marks
1	Perform the Design, Development and Simulation of a network according to Three Tier architecture 1.1 Design network diagram using packet tracer 1.2 Design proper IP addressing with VLSM. 1.3 Configure all devices with IP addressing.	2	3
2	Perform the Design, Develop and Simulation of Login using Telnet and SSH. 2.1 Design network diagram using packet tracer 2.2 Design proper IP address with network devices. 2.3 Configure remote access (Telnet, SSH) 2.4 Ensure Login operation by username and password.	1	3
3	Perform the Design, Develop and Simulation of DHCP server 3.1 Design network diagram for DHCP service (such as – in Cisco & Mikrotik) using packet tracer/GNS3 3.2 Design proper IP address with network devices. 3.3 Configure DHCP server 3.4 Ensure all clients get IP information	1	3
4	Perform the Design, Develop and Simulation of Static & Default Static Routing 4.1 Design network diagram using packet tracer 4.2 Design proper IP address with network devices 4.3 Configure Static Routing and default static routing 4.4 Ensure reachability	1	3
5	Perform the Design, Develop and Simulation of Open Shortest Path Fast Routing Protocol(OSPF) 5.1 Design network diagram using packet tracer 5.2 Design proper IP address with network devices 5.3 Configure OSPF 5.4 Ensure reachability	1	3
6	Perform the design & implementation of Domain Controller 6.1 Install the Windows Server 2022 into VM-Ware / Hyper-V / Oracle Virtual Box 6.2 Install the Active Directory Domain Service into Windows Server 2022 6.3 Configuring the Active Directory Domain Service 6.4 Join the client machines into domain 6.5 Create the domain users into server & Login by domain user from client machines	1	3
7	Deployment of Group Policy Object(GPO) 8.1 Create and apply group policy 8.2 Create local group policy 8.3 Create domain group policy 8.4 Create organizational unit(OU) group policy 8.5 Create security group policy 8.6 Practice on configuring of group policies (Software installation, Internet explorer maintenance, Controlling USB devices, Restricting user accounts and logging times etc.)	2	7
8	Perform the Design, Develop and Simulation of Hot Standard Router Protocol (HSRP) 9.1 Design network diagram using packet tracer. 9.2 Design proper IP address with network devices. 9.3 Configure HSRP 9.4 Ensure reachability.	1	3

9	Perform the Design, Develop and Simulation of Network Address Translator (NAT) 10.1 Design network diagram using packet tracer 10.2 Design proper IP address with network devices 10.3 Configure NAT as per requirement 10.4 Ensure reachability	1	3
10	Perform the Design, Develop and Simulation of Standard Access Control List(ACL) 11.1 Design network diagram using packet tracer 11.2 Design proper IP address with network devices 11.3 Configure ACL 11.4 Ensure controlling the access	1	3
11	Perform the Design, Develop and Simulation of Port Security in Switch 14.1 Design network diagram using packet tracer 14.2 Design proper IP address with network devices. 14.3 Configure Port Security 14.4 Check configuration	1	3
12	Perform the Design, Develop and Simulation of Inter- VLAN Routing 15.1 Design network diagram using packet tracer 15.2 Design proper IP address with network devices. 15.3 Create VLAN into all switches. 15.4 Configure the inter-VLAN routing 15.5 Ensure configuration	1	3
	Total	16	50

NECESSARY RESOURCES (TOOLS, EQUIPMENT AND MACHINERY):

SL	Item Name	Quantity
1	Network tool box	10 nos
2	Cat6 and Fiber Optic cable	5 box
3	RJ 45 connector	300 nos
4	Patch Cord and SFP Module	10 nos
5	Splicer Machine	5 nos
6	Crimping tool	10 nos
7	Cable tester	10 nos
8	Hub	05 nos
9	Repeater	10 nos
10	Network Switch-8 Ports	5 nos
11	Network Switch- 24 Ports	5 nos
12	Network Switch-48 Ports	5 nos
13	Router (Wireless)	10 nos
14	Cisco Router	05 nos
15	Mikrotik Router (5P, 8P, 16P)	10 nos
16	Brouter	5 nos
17	Desktop PC	20 nos
18	Laser Printer	5 nos
19	Internet connection	At least 20Mbps
20	Windows server 2012/2019/2022 DVD(Licensed)	10 nos

RECOMMENDED BOOKS:

SL	Book Name	Writer Name	Publisher Name & Edition
01.	Cisco Certificate Network Associate study Guide	Todd Lammle	Sybex.
02.	Computer Networks	Andrew S.Tanenbaum	PRENTICE HALL, 5th Edition
03.	Data Communication &Networking	YEKINI N. ASAFE ADEBARI F. ADEBAYO BELLO OLALEKAN	Computer Engineering Department Yaba College ofTechnology Lagos Nigeria
04.	Official CertificateGuide	Wendell Odom	Cisco press
05.	Cisco Certificate Network Associate study Guide	Todd Lammle	Sybex.
05	Data Communications and Networking	Behrouz A. Forouzan	McGraw Hill 5th Edition

WEBSITE REFERENCES:

SL	Web Link	Remarks
1	(a) Website: https://en.wikipedia.org/wiki/Data_communication (b) Website: https://en.wikipedia.org/wiki/Computer_network (c) Website: https://en.wikipedia.org/wiki/Windows_Server (d) Website: https://en.wikipedia.org/wiki/Cloud_storage (e) Website: https://en.wikipedia.org/wiki/Cloud_computing	Search here with given link
2	www.youtube.com	Search here with topics
3	www.google.com	Search here with topics

Subject Code	Subject Name	Period/Week		Credit
28573	Cyber Security & Ethics	T	P	C
		2	3	3

Rationale	This is an occupational-specific subject in the curriculum for diploma in engineering courses required to enable the graduate to use and work with Cyber security field which is very essential for an IT professional. This course enhances the students' knowledge & skills on Cyber security fields and also help to provide job placement or self-employment of the students.
Learning Outcome (Theoretical)	After completing the theoretical course, students should be able to: • interpret types and methodology of Cyber security & Risk management, importance of Cyber Security Laws & rules • illustrate of computer & Internet security, security resources, security tools, types of Firewall, types of threat & attack and securing data storage system • explain details of Vulnerability Assessment, Penetration Testing • describe basic of Blockchain technology, Data Security and Digital Forensics • explain about Incidents Response and Disaster Recovery • illustrate basics of IT Security Auditing
Learning Outcome (Practical)	After completing the practical course, students should be able to: • apply network security, system security and basic data security • perform the techniques of exploit vulnerabilities and techniques of post exploitation of vulnerabilities • apply data hiding techniques using Steganography and Cryptography techniques • apply application data security • perform system forensics, mobile and IoT forensics • prepare forensic report • apply the Governance and Management of IT audit

Detailed Syllabus (Theory)

Unit	Topics with Contents	Period	Marks
1	Cyber Security and Risk Management 1.1 Define cyber security & goal of cyber security 1.2 Interpret cyber security & risk management 1.3 Explain structure of cyber world 1.4 Illustrate type of cyber security 1.5 Explain functional framework of cyber security 1.6 Describe risk management framework 1.7 Illustrate basic risk analysis methodology & risk mitigation process 1.8 Illustrate ethical hacking 1.9 Interpret importance of cyber security laws & rules 1.10 Explain user awareness and training for cyber security	3	10
2	System and Network Security 2.1 State system and network security 2.2 Describe computer, network & internet security resources 2.3 Explain the use of security tools 2.4 Define firewall & state the types of firewall 2.5 Define threat and mention types of threat 2.6 Illustrate wireless attack & attacks on different layer 2.7 Illustrate basic data security and securing data storage system	3	8
3	Vulnerability Assessment and Penetration Testing 3.1 Define vulnerability assessment and penetration testing (VAPT) 3.2 Describe types of vulnerability scanning & penetration testing 3.3 explain penetration testing phases and techniques 3.4 Illustrate wireless hacking and different types of hacking tools 3.5 Explain log files and command & control Centre (CnC) 3.6 State VAPT report	4	10
4	Data Security 4.1 Define data security, database security and data privacy 4.2 Illustrate different types of steganography 4.3 Describe various data hiding techniques 4.4 Explain steganography and steganography algorithms 4.5 Explain data encryption algorithms and techniques 4.6 Explain encrypted data storage technique is used in databases 4.7 Illustrate digital signatures and digital certificates techniques	5	8
5	Blockchain Technology 5.1 State blockchain & blockchain technology 5.2 Mention application field of blockchain technology 5.3 Explain blockchain mining process 5.4 Describe blockchain security techniques	3	5
6	Digital Forensics 6.1 Define digital crime 6.2 Mention the classification of digital crime	6	8

	6.3 Illustrate digital forensic and digital forensic investigation processes 6.4 State the type of file system and file carving techniques 6.5 Describe the techniques of network log forensic, cloud forensic techniques, database forensic techniques, network log forensic technique, cloud forensic techniques, malware forensic and mobile forensic. 6.6 Explain defeating anti-forensic techniques is applied. 6.7 Describe forensic investigation report		
7	Incidents Response and Disaster Recovery 7.1 Define incident handling and disaster recovery 7.2 Describe the classification of incident elements handling process 7.3 Explain phases of incident handling 7.4 Illustrate disaster recovery strategy and policy 7.5 Explain steps of disaster recovery	4	6
8	IT Security Auditing 8.1 Define it audit 8.2 Describe types of it audit 8.3 Explain the process of auditing information systems 8.4 Explain the processes of it auditing 8.5 Explain it audit sampling methodology 8.6 Describe it audit reporting process	4	5
	Total	32	60

Detailed Syllabus (Practical)

Sl.	Experiment name with procedure	Class (3 Period)	Marks
1	Apply network security 1.1 Identify Network security tools as per requirement 1.2 Implement the Network firewall configuration as per job sheet 1.3 Apply Wireless security as per Job Sheet of course Instructor 1.4 Submit records to the course instructor.	1	5
2	Apply system security 2.1 Identify System security tools as per requirement 2.2 Apply the System hardening as per instruction sheet (supplied by the course instructor) 2.3 Apply basic System commands for system security as per instruction sheet of instructor 2.4 Apply Host based security as per Job sheet 2.5 Implement the System firewall configuration as per job sheet 2.6 Submit records to the course instructor	1	5
3	Apply basic data security 3.1 Identify basic data security & data encryption tools as per requirement 3.2 Implement Strong password mechanism for data security 3.3 Apply Data Encryption using encryption tools	1	4

	3.4 Submit records to the course instructor		
4	Apply the techniques of exploit vulnerabilities 4.1 Identify basic System hacking tools as per requirement 4.2 Apply System hacking tools as per instruction sheet 4.3 Use Sniffing tools as per instruction sheet 4.4 Use Malware analysis tools as per instruction sheet 4.5 Apply Wireless hacking tools as per instruction sheet 4.6 Use Mobile device / platform hacking tools as per instruction sheet 4.7 Use Web server and web application hacking tools as per instruction sheet	2	4
5	Perform the techniques of post exploitation of vulnerabilities 5.1 Review the Vulnerabilities assessment 5.2 Perform disabling protections 5.3 Apply Local vulnerabilities Assessment 5.4 Perform creation and Leveraging of Backdoors 5.5 Manage Log files 5.6 Apply Data search and extraction techniques 5.7 Generate VAPT Report	3	4
6	Apply data hiding techniques using Steganography 6.1 Identify Steganography detecting tools as per requirement 6.2 Apply Streaming media 6.3 Apply multilingual steganography	1	4
7	Apply Cryptography techniques 7.1 Identify required tools as per job sheet supplied by the course instructor 7.2 Use Digital signatures and digital certificates techniques 7.3 Apply Cryptosystems as per instruction sheet 7.4 Use Hash functions as per requirement	1	4
8	Apply application data security 8.1 Identify required tools as per job sheet supplied by the course instructor 8.2 Use Encrypted data storage technique in databases. 8.3 Rectify SQL and Code injection attacks 8.4 Apply Encrypted network protocols for data security.	1	4
9	Perform system forensics 9.1 Identify required tools as per job sheet supplied by the course instructor 9.2 Perform Network log forensic technique 9.3 Perform Cloud forensic techniques and database forensic techniques are performed. 9.4 Apply Malware forensic technique 9.5 Apply Defeating anti-forensic techniques 9.6 Perform Operating system forensics	2	4
10	Perform mobile and IoT forensics 10.1 Identify required tools as per job sheet supplied by the course	1	4

	instructor		
	10.2 Apply Mobile forensic process is defined.		
	10.3 Perform IoT forensic on IoT devices		
11	Prepare forensic report 11.1 Identify required tools as per job sheet supplied by the course instructor 11.2 Prepare Forensic investigation report template 11.3 Prepare Chain-of-Custody document 11.4 Prepare Expert witness report 11.5 Maintain Legal procedure 11.6 Prepare final report	1	4
12	Apply the Governance and Management of IT audit 12.1 Identify required tools as per job sheet supplied by the course instructor 12.2 Perform Business Impact Analysis (BIA) 12.3 Prepare Business Continuity Plan (BCP) 12.4 Performed IT Audit according to IT governance and management practices	1	4
	Total	16	50

Necessary Resources (Tools, equipment's and Machinery):

Sl	Item Name	Quantity
01	Desktop PC	25 nos
02	Internet connection	At least 20Mbps

Recommended Books:

Sl	Book Name	Writer Name	Publisher Name & Edition
01	Cybersecurity Ethics and Introduction	By Mary Manjikian	1st Edition
02	The Ethics of Cybersecurity	Christen, Markus (editor) Gordijn, Bert (editor) Loi, Michele (editor)	
03	Cybersecurity, Ethics, and Collective Responsibility	Seumas Miller and Terry Bossomaier	

Website References:

Sl	Web Link	Remarks
01	http://library.oapen.org/handle/20.500.12657/22489	
02	https://www.ollusa.edu/blog/cybersecurity-ethics.html	

Subject Code	Subject Name	Period Per Week		Credit
28574	APPS DEVELOPMENT PROJECT	T	P	C
		1	3	2

Rationale	By engaging in an Android app development learning project, students can build a strong foundation in mobile development, enhance their technical skills, and open up a wide range of career and entrepreneurial opportunities. Nevertheless, the course focuses on Android development involves mastering programming languages like Java and Kotlin, enhancing overall coding skills, Android development opens avenues for freelance work and consulting projects, mobile development best practices, UI/UX design, and performance optimization, Android app development skills can be applied to various types of applications, from games and productivity tools to health and educational apps, understanding cross-platform frameworks like Flutter, which enable the creation of apps for both Android and iOS, Developing apps involves solving complex problems and debugging, which enhances analytical and problem-solving skills, latest trends and technologies in mobile development, such as AI, machine learning and IoT and software design patterns, architecture and efficient coding practices.
Learning Outcome (Theoretical)	After completing the subject, students will be able to: 1. Interpret Android Environment. 2. Explain Activities and Fragment of Android. 3. State the procedure to Create Android User Interface. 4. Elucidate Android Intents, Filters and Navigation. 5. Explain Data Storage Management. 6. Illustrate Networking and Web Services for Android. 7. Explain Advanced Android Features. 8. State the process of Publish the App.
Learning Outcome (Practical)	After undergoing the subject, students will be able to: 1. Create Android Environment for App Development 2. Perform Kotlin Programming 3. Create Android User Interface. 4. Demonstrate Activities & Fragment 5. Demonstrate Android Intents and Filters 6. Operate on Data Storage Management 7. Operate RESTful API and Web Services 8. Perform Content Providers 9. Perform Location Based Services of Android 10. Publish the App

Detailed Syllabus (Theory)

Unit	Topics with contents	Class (Period)	Final Marks
1	Android and Kotlin Environment 1.1 State Android Operating System. 1.2 Describe Kotlin Programming Language. 1.3 Explain OOP in Kotlin. 1.4 Illustrate Android Application Project Structure with example.	2	5
2	Activities and Fragment 2.1 State Activity Lifecycle. 2.2 Illustrate Android application 2.3 Describe the activity lifecycle methods 2.4 Explain Fragment Lifecycle and Use 2.5 Explain Types of Fragments.	2	5
3	Android User Interface 3.1 Explain XML Layout and Programmatic Layouts. 3.2 State the Basic Building Block for User Interface. 3.3 Describe the process to Use Image View, Image Switcher to Display Pictures. 3.4 Explain Frame Layout, Linear Layout, Table Layout, Relative Layout, List View, Grid View, Recycler View. 3.5 Illustrate Jetpack Compose. 3.6 State Basic Composable and Layouts. 3.7 State Management in Compose. 3.8 Explain Jetpack Compose Project Setup. 3.9 Differentiate between XML Layout and Programmatic Layouts.	2	5
4	Android Intents, Filters and Navigation 4.1 Illustrate Android Intents and Filters 4.2 State the Intent Object and Fundamental Use Cases. 4.3 Explain Intent Types. 4.4 State Data Transfer between Activities. 4.5 Illustrate Navigation	2	5
5	Data Storage Management 5.1 Describe Device Storage: Internal Storage, External Storage. 5.2 Explain Shared Preferences for Simple Data Storage. 5.3 Illustrate Storing and Retrieving Data with Shared Preferences. 5.4 State Room Persistence Library.	2	5
6	RESTful API and Web Services 6.1 Describe Networking Basics in Android. 6.2 State RESTful API. 6.3 Explain Retrofit for API Calls. 6.4 Illustrate RESTful APIs with Retrofit. 6.5 State JSON Parsing and Data Binding.	2	5
	TOTAL	12	30

Detailed Syllabus (Practical)

Sl. No.	Experiment name with procedure	Class (3 Period)	Continuous Marks
1	Create Android Environment for App Development 1.1 Install the latest Android Studio. 1.2 Install Android SDK. 1.3 Configure and Explore the IDE. 1.4 Create Hello! World Application. 1.5 Inspect Android Application Project Structure (Folder, File & Description, Manifest File, Strings File, Layout File). 1.6 Configure and run App on Emulator and Real Device.	1	2
2	Perform Kotlin Programming 2.1 Perform with Variables, Comments, Data Types, Branching & Looping, Function, Class & Objects, Inheritance, Constructor, Abstract, Collection and Arrays. 2.2 Operate with Kotlin Standard Library (String Manipulation and Null Safety, Nullable types and safe calls, Extension. 2.3 Perform Higher-order Function and Lambdas, Elvis operator. 2.4 Perform with Exception Handling and Debugging.	2	3
3	Create Android User Interface. 3.1 Integrate basic building block for User Interface. 3.2 Add Text, Buttons, Checkboxes, Radio Buttons, Toggle Buttons, Spinners, Pickers, Tooltips to the layout files. 3.3 Adapt layout attributes. 3.4 Construct Frame layout, Linear layout, Table layout, Relative layout, List View, Grid View. 3.5 Set Up a Jetpack Compose Project. 3.6 Build Simple UIs with Compose. 3.7 Handle State and Interactions in Compose. 3.8 Use Image View, Image Switcher to Display Pictures. 3.9 Use Grid View to Display Pictures. 3.10 Manipulate Menus with Views.	2	3
4	Demonstrate Activities & Fragment of Android 4.1 Create an Android Application and test activity lifecycle your app's activities. 4.2 Create a new activity and perform show 'Hello! World' text filed on Click Event using XML and Jetpack Compose. 4.3 Create a Simple Calculator to perform Addition, Subtraction, Multiplication and Division. 4.1 Create a Fragment Class. 4.2 Add a Fragment to an Activity at Runtime. 4.3 Replace One Fragment with Another. 4.4 Communicate with other Fragments. 4.5 Deliver a Message to a Fragment.	1	3

Sl. No.	Experiment name with procedure	Class (3 Period)	Continuous Marks
5	Demonstrate Android Intents and Filters 5.1 Perform different use cases on Intent object and fundamental. 5.2 Manipulate Intent types and Data transfer between activities. 5.3 Register an Activity for the Intent which is triggered when someone wants to open a webpage. 5.4 Design effective navigation. 5.5 Create a swipe view with tabs. 5.6 Create a navigation drawer. 5.7 Implement proper Back navigation.	1	2
6	Operate on Data Storage Management 6.1 Save Files on Device Storage - Internal Storage, External Storage. 6.2 Use Shared Preferences to store User Name after Login. 6.3 Store and Retrieve Data with Shared Preferences. 6.4 Create an app to perform CRUDE (Create, Retrieve, Update, Delete) operation using Room Database.	1	2
7	Operate with RESTful API and Web Services 7.1 Implement GET, POST, PUT, PATCH, DELETE with Retrofit/Volley. 7.2 Perform JSON Parsing and Data Binding. 7.3 Create an app respond to to check Device network Connection. 7.4 Create an app to perform CRUDE (Create, Retrieve, Update, Delete) operation using Firebase.	1	2
8	Apply Content Providers of Android 8.1 Create an address book using the Content Provider. 8.2 Maintain the record of performed task.	1	2
9	Perform Location Based Services of Android 9.1 Project– Build a Location Tracker. 9.2 Display Maps. 9.3 Get the current location. 9.4 Get the Updated Location. 9.5 Displaying a Location Address.	1	2
10	Publish App 10.1 Prepare for release, gather materials and resources. 10.2 Set Application version information. 10.3 Create Certificates and KeyStore, Manage keys. 10.4 Upload your App Bundles. 10.5 Update your App.	1	2
11.	Any Two Experiment of the Six Experiment should be done 11.1 Work with Messaging. - Understand SMS Manager API. - Analyze Method & Description. - Send an SMS to the given mobile number. - Analyze Email Messaging. - Analyze Extra Data & Description.	4	2

Sl. No.	Experiment name with procedure		Class (3 Period)	Continuous Marks
		Send an Email to the given recipients.		
	11.2	Develop Android Services - Creating Own Services - Long Running Task and Services, Repeated Task in a Service, Execute a synchronous Task on separate. - Establishing Communication between a Service and Activity. - Bind activities to services. - Understand Threading.		
	11.3	Configure your build - The build processes. - Custom build configurations. - Build configuration files. - Set the application ID. - Add build dependencies, Dependency types, configurations. - Optimize your build speed. - Configure build variants.		
	11.4	Create a Contact Book. - Understand the project scope and requirement analysis. - Create a Form that will take username, gender, email, phone number. - Upload user image. - Store data to database. - Update and Delete a contact. - Display user information in List View. - Test your App with UI-Testing and with testing tools.		
	11.5	Create a BMI Tracker - Understand the project scope and requirement analysis. - Create a calculator that will take user height, weight and age. - Create BMI Standard to compare. - Show user BMI and give status. - Store BMI result. - Display monthly BMI. - Test your App with UI-Testing and with testing tools.		
	11.6	Create a Location based App. - Understand the project scope and requirement analysis - Create a form that will take name of type like Hospital, Institute, Fire station. - Create a form that will take information about Address, contact number, latitude, longitude of		

Sl. No.	Experiment name with procedure		Class (3 Period)	Continuous Marks
		the address, basic service-related information. • Create a signup form for user and store. • Filter nearby locations stored in database. • Test your App with UI-Testing and with testing tools.		

SL.	Book Name	Writer Name	Publisher Name & Edition
01	Beginning Android Programming with AndroidStudio	J F DIMarjio	
02	Sams Teach Yourself - Android Application Development in 24 Hours	Shane Conder	3 rd Edition

Website References:

SL. No.	Web Link	Remarks
01	https://developer.android.com/	
02	https://www.tutorialspoint.com/android	
03	https://www.geeksforgeeks.org/introduction-to-android-development/	

Subject Code	Subject Name	Period per Week		Credit
28575	Multimedia and Animation	T	P	C
		2	3	3

Rationale	The course prepares students for careers in multimedia design, development, and analysis by providing a blend of theoretical knowledge and practical skills, ensuring they are well-equipped to meet industry demands and innovate in the field of multimedia.
Learning Outcome (Theoretical)	1. Explain the history and key differences between animation and multimedia. 2. State applications of animation and multimedia across various industries. 3. Analyze the animation and multimedia production pipeline. 4. Explain Application of design principles (balance, contrast, hierarchy, etc.) and color theory to create visually appealing compositions. 5. Illustrate storytelling techniques for effective communication in animation and multimedia projects. 6. Explain the core principles of animation (squash & stretch, anticipation, follow-through) and their impact on movement. 7. Design visually engaging characters through an understanding of character design fundamentals. 8. Explain the importance of timing and spacing for creating believable animation. 9. Apply motion graphics design principles (typography, layout, animation) to projects. 10. Navigate the 3D environment and understand 3D space and coordinate systems. 11. Explain different 3D modeling techniques (polygonal modeling, sculpting, etc.). 12. State the Application of texturing and materials to enhance the realism of 3D models. 13. State the principles of rigging and character animation for 3D projects. 14. Explain compositing concepts like layering, masking, and green screen keying. 15. Utilize color correction and grading techniques to achieve a desired visual style. 16. Apply video editing theory (cuts, transitions, pacing) for storytelling purposes. 17. Interpret VFX (particle effects, rotoscoping, compositing VFX). 18. Explain sound design fundamentals (foley, dialogue, sound effects) and their role in animation and storytelling. 19. State audio editing techniques (cutting, fading, layering, effects) for effective sound design. 20. Describe the process to Synchronize sound with animation to create a cohesive experience. 21. Compare and contrast freelancing vs. in-house work in the animation and multimedia industry. 22. Develop strategies for building a strong portfolio and showcasing their skills. 23. Explain effectively marketing techniques and network within the industry. 24. Interpret copyright and licensing considerations for animation and multimedia projects. 25. Analyze successful animation and multimedia projects to identify design principles and techniques. 26. State the procedure to constructive critiques of their own and others' work.

	27. Elucidate future trends in animation and multimedia design.
Learning Outcome (Practical)	1. Design logos, branding elements, and other graphic design projects using design principles and software. 2. Animate a short scene featuring a bouncing ball with proper adherence to animation principles. 3. Create a short animated explainer video using motion graphics techniques. 4. Model and animate a simple 3D object. 5. Composite a 3D animation project with a background and add visual effects. 6. Enhance an animation project with basic particle effects. 7. Design and edit sound effects and music to create a cohesive soundscape for an animation project. 8. Develop a final animation project showcasing all the learned skills, including 2D animation, 3D animation (optional), VFX, sound design, and editing. 9. Build a professional portfolio to effectively showcase their animation and multimedia skills.

Detailed Syllabus (Theory)

Unit	Topics with contents	Class (Hours)	Final Marks
1	Introduction to Multimedia and Animation 1.1 State the history of Multimedia & Animation. 1.2 Mention the types of Multimedia & Animation 1.3 Differentiate between Multimedia & Animation 1.4 Describe the Applications of Multimedia & Animation for film, advertising, games. 1.5 Explain 2D and 3D Animation Production Pipeline	2	6
2	Design Principles & Visual Communication 2.1 Mention the Elements of design 2.2 State Design Principles in Depth 2.3 Explain Color Theory & Psychology 2.4 Interpret Visual Perception & Composition 2.5 Storytelling Techniques for Multimedia & Animation	4	6
3	2D Animation 3.1 Illustrate Animation Principles 3.2 State Character Design Fundamentals 3.3 Explain Timing & Spacing in Animation	4	8

4	3D Animation 4.1 Explain 3D Animation production Pipeline 4.2 State 3D Space & Coordinate Systems 4.3 Explain 3D Modeling Techniques 4.4 Illustrate Lighting, Texturing & Materials with adding realism to 3D models. 4.5 Explain Rigging & Character Animation for 3D with principles of character setup and animation.	4	8
5	Motion Graphics 5.1 Define Motion graphics. 5.2 State the applications of Motion graphics. 5.3 Describe the After Effects of Motion Graphics 5.4 State the interface & core features of Motion graphics 5.5 Illustrate importing Assets (basics) & Hands-on: Simple Animation (using keyframes) 5.6 Explain Design Principles for Motion Graphics (overview)	4	8
6	VFX, Compositing & Editing Theory 6.1 State Compositing Concepts 6.2 Explain layering, masking and green screen keying. 6.3 Illustrate Color Correction & Grading Techniques for enhancing the visual style. 6.4 Interpret Video Editing Theory. 6.5 State VFX Concepts with particle effects, rotoscoping and compositing VFX.	4	8
7	Sound Design & Audio Theory 7.1 State Sound Design Fundamentals with foley, dialogue and sound effects. 7.2 Describe the Importance of Sound in Animation & Storytelling 7.3 Explain Audio Editing Techniques with cutting, fading, layering and effects. 7.4 State Synchronization Techniques with matching sound with animation.	4	6
8	Business of Multimedia & Animation 8.1 Explain Freelancing and In-house Work in the Industry 8.2 State the procedure to create a Portfolio & Showcasing Your Skills 8.3 Explain the technique for Marketing Yourself & Networking in the Industry 8.4 Elucidate the Copyright & Licensing for Multimedia & Animation Projects	2	4
9	AI for Animation and Video Editing Tools 9.1 State AI-powered tools. 9.2 Explain AI-assisted video editing.	4	6

	9.3 State the uses of AI tools to enhance a short animation or video clip (provided or student-made).		
	Total	32	60

Detailed Syllabus (Practical)

SL.	Experiment with working procedure	Class (Period)	Marks (Continuous)
1.	Observe of Graphic Design Software 1.1 Analyze Design Principles (balance, contrast, hierarchy) 1.2 Identify Color Theory (color psychology, creating palettes) 1.3 Practice Typography (choosing fonts, hierarchy, readability) 1.4 Select Design Software (such as Adobe Photoshop/ Adobe 1.5Illustrator or any other design software) 1.5 Maintain the record of performed task.	2	4
2.	Develop 2D Digital Animation 2.1 Identify Animation Principles (squash & stretch, anticipation, follow-through) 2.2 Identify Storyboarding and creating animatics 2.3 Traditional animation techniques (exposure sheets, flipbooks) 2.4 Introduction to 2D animation software (Adobe Animate or similar) 2.5 Character Design and Animation Basics	2	2
3.	Create 2D Motion Graphics 3.1 Observe Motion Graphics principles (typography in motion, visual hierarchy) 3.2 Perform designing for motion (creating assets for animation) 3.3 Work with keyframes and timelines in animation software 3.4 Animate text, shapes, and logos for motion graphics projects	2	2
4.	Perform 3D Modeling & Animation 4.1 Perform 3D space navigation and manipulating objects 4.2 Apply Perform Basic 3D modeling techniques (polygon modeling, sculpting) 4.3 Perform Lighting, Texturing and applying materials to 3D models 4.4 Perform Rigging and character setup for animation 4.5 Create simple 3D animation exercises (bouncing ball, character walk cycle)	2	3

5.	Perform Compositing & Editing 5.1 Perform compositing software (Adobe After Effects or similar) 5.2 Build with layers and masks in compositing 5.3 Operate Green screen keying basics 5.4 Operate Basic color correction and effects 5.5 Perform video editing concepts (cuts, transitions, pacing)	1	2								
6.	Create Visual Effects (VFX) for Animation 6.1 Observe VFX concepts (particle effects, rotoscoping) 6.2 Create basic particle effects (fire, smoke, explosions) 6.3 Implement Rotoscoping techniques (isolating objects from live-action footage) 6.4 Apply Integrating VFX elements into your animation projects	1	2								
7.	Perform Sound Design & Audio Editing 7.1 Identify Sound libraries and finding sound effects 7.2 Design sound for animation (foley, dialogue) 7.3 Perform Basic audio editing techniques (cutting, fading, layering) 7.4 Synchronize sound with animation	2	2								
8.	Develop Project & Portfolio Building 8.1 Develop a final project concept using all learned skills (animation, VFX, sound design) 8.2 Perform Storyboarding, animation, and editing for the final project 8.3 Prepare portfolio presentation techniques to showcase your work Course wrap-up and career exploration	2	4								
9.	<u>Any One of the four should be done</u> <table><tr><td>9.1</td><td>Project: Design a logo and branding elements for a fictional company.</td></tr><tr><td>9.2</td><td>Project: Animate a short scene featuring a bouncing ball interacting with its environment, adhering to animation principles.</td></tr><tr><td>9.3</td><td>Project: Create a short animated explainer video using motion graphics techniques to explain a simple process.</td></tr><tr><td>9.4</td><td>Project: Model and animate a simple 3D object (e.g.,) bouncing ball with tail) in a scene.</td></tr></table>	9.1	Project: Design a logo and branding elements for a fictional company.	9.2	Project: Animate a short scene featuring a bouncing ball interacting with its environment, adhering to animation principles.	9.3	Project: Create a short animated explainer video using motion graphics techniques to explain a simple process.	9.4	Project: Model and animate a simple 3D object (e.g.,) bouncing ball with tail) in a scene.	1	2
9.1	Project: Design a logo and branding elements for a fictional company.										
9.2	Project: Animate a short scene featuring a bouncing ball interacting with its environment, adhering to animation principles.										
9.3	Project: Create a short animated explainer video using motion graphics techniques to explain a simple process.										
9.4	Project: Model and animate a simple 3D object (e.g.,) bouncing ball with tail) in a scene.										
10.	<u>Any One of the four should be done</u> <table><tr><td>10.1</td><td>Project: Composite your 3D animation project into a scene with a background and add visual effects (e.g., lighting, shadows).</td></tr></table>	10.1	Project: Composite your 3D animation project into a scene with a background and add visual effects (e.g., lighting, shadows).	1	2						
10.1	Project: Composite your 3D animation project into a scene with a background and add visual effects (e.g., lighting, shadows).										

	10.2	Project: Enhance project by adding basic particle effects (e.g., smoke trail behind the bouncing object).		
	10.3	Project: Add sound design and audio editing to project, creating a cohesive soundscape that complements the visuals.		
	10.4	Project: Create a short animated video incorporating all the techniques learned throughout the course, including 2D animation, 3D animation (optional), VFX, sound design, and editing.		
	Total		16	25

RECOMMENDED BOOKS:

SL	Book Name	Writer Name	Publisher Name & Edition
	References:	References:	References:
01.	Animator's Survival Kit	Richard Williams (Third Edition). (2009)	Focal Press.
02.	Substance, Structure, Style and the Principles of Screenwriting	McKee, Robert. (1997)	HarperCollins Publishers.
03.	The Non-Designer's Design Book (Fourth Edition). (2015).	Robin Williams	Peachpit Press.
04.	3D World Magazine: Essential Techniques for 3D Artists (Third Edition).	Watt, Andrew. (2016).	3D World

WEBSITE REFERENCES:

SL	Web Link	Remarks
1	https://color.adobe.com/create/color-wheel	
2	https://www.blender.org	
3	https://freesound.org	
4	https://audiojungle.net	
5	https://www.awn.com	
6	https://books.google.com.bd/books?id=3AQFN4IzBAoC&printsec=frontcover&redir_esc=y#v=onepage&q&f=false	
7	https://www.shiksha.com/animation-syllabus-chp	
8	https://www.pacificu.edu/multimedia/multimedia-courses	

Subject Code	Subject Name	Period/Week		Credit
28576	Project Works-II	T	P	C
		0	6	2

Rationale	The subject Project Works-II aims to equip students with practical skills in modern technology applications. By developing an information management system, students will learn systematic data handling. Establishing and configuring network systems will enhance their understanding of connectivity. Developing IoT systems will prepare them for the future of interconnected devices. Designing and developing surveillance security systems will address the growing need for safety solutions. Finally, creating e-commerce sites will provide insights into the digital marketplace, essential for today's economy. These projects collectively ensure comprehensive exposure to critical areas in computer science and technology, fostering innovation and technical proficiency.
Learning Outcome (Practical)	After undergoing the subject, students will be able to: 1. Develop Information management system methodically 2. Establish and configure Network system 3. Develop IoT system 4. Design and Develop surveillance Security system 5. Design and develop e-commerce site

(Practical)-Project based (Out of 8 minimum 2 project must be completed)

Sl	Project Name (Please choose any 2 of them)	Remarks
1	Design and Develop a Library Management System / Similar any project designed / approved by the guide teacher.	
2	Project Title: Design and develop a physical / simulation network /using physical network device / Cisco Packet Tracer / Similar any project designed / approved by the guide teacher.	
3	Project Title: Design and develop a Network Traffic Analyzer / Similar any project designed / approved by the guide teacher.	
4	Project Title: Design and develop an Automated Plant Watering System / Similar any project designed / approved by the guide teacher.	
5	Project Title: Design and develop a Smart Home System / Similar any project designed / approved by the guide teacher.	
6	Project Title: Design and develop a License Plate Recognition (LPR) System/ Similar any project designed / approved by the guide teacher.	
7	Project Title: Design and develop an Online Course Management System / Similar any project designed / approved by the guide teacher.	
8	Project Title: Create a basic e-commerce application using Java / Similar any project designed / approved by the guide teacher.	

Detailed Syllabus (Practical)

Sl.	Experiment name with procedure	Class (3 period)	Continuous Marks
1	Project Title: Design and Develop a Library Management System / Similar any project designed / approved by the guide teacher. Project Requirements: - Software: MySQL or PostgreSQL, a front-end tool like PHPMyAdmin or MySQL Workbench - Basic knowledge of SQL and database design Steps of Activities:	16	25
	1. Requirements Gathering: Identify the entities (books, borrowers, transactions) and their attributes	1	2
	2. Database Design: Create an E-R diagram and define the relationships between entities.	2	3
	3. Schema Creation: Write SQL scripts to create tables and define relationships	4	5
	4. Data Entry: Insert sample data into tables.	1	2
	5. Queries: Develop queries for common operations like searching books, checking out books, and returning books.	4	5
	6. Front-End Integration: Optional - Create a simple front-end interface to interact with the database using a web development technology.	2	4
	7. Presentation: Prepare presentation on the project activities and disseminate during the final exam to the internal and external examiner / assessor.	2	4

	Guiding Principles: • Data normalization: Minimize redundancy and ensure data integrity. • Secure user authentication: Use strong password hashing techniques. • Transaction management: Maintain data consistency during concurrent access. • Data validation: Ensure input accuracy to prevent invalid data. • Backups and recovery: Regularly back up the database for disaster recovery.
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Sl.	Experiment name with procedure	Class (3 period)	Marks
2	Project Title: Design and develop a physical / simulation network /using physical network device / Cisco Packet Tracer / Similar any project designed / approved by the guide teacher. Project Requirements: • Software: Cisco Packet Tracer / GNS 3 / similar any software • Basic understanding of networking concepts	16	25
	Steps of Activities: 1. Project Planning: Define the network topology, including devices like routers, switches, and workstations.	2	3
	2. Topology Setup: Use Cisco Packet Tracer / GNS-3 to arrange the devices according to the planned topology.	2	3
	3. Configuration: Configure IP addresses, subnets, and VLANs.	3	5
	4. Routing Protocols: Implement routing protocols (e.g. OSPF, EIGRP) to enable communication between different subnets.	2	3
	5. Testing and Validation: Test the network by pinging devices and checking connectivity.	4	5
	6. Documentation: Document the network design and configuration steps.	1	3
	7. Presentation: Prepare presentation on the project activities and disseminate during the final exam to the internal and external examiner / assessor.	2	3
	Guiding Principles: • Network neutrality: Respect network traffic without compromising privacy. • Ethical considerations: Obtain necessary permissions before capturing network traffic. • Network performance impact: Minimize the tool's impact on network performance. • Data security: Securely store and analyze captured traffic data.		

Sl.	Experiment name with procedure	Class (3 period)	Marks
3	Project Title: Design and develop a Network Traffic Analyzer / Similar any project designed / approved by the guide teacher. Project Requirements: • Capture and analyze network traffic in real-time. • Identify different network protocols (TCP, UDP, IP, etc.). • Display network traffic statistics (bandwidth usage, packet types, top sources/destinations).	16	25

	• Detect potential security threats (malicious traffic patterns, anomalous activity). • Generate reports on network performance and usage trends.		
	1. Technology Selection: Choose a network traffic capture tool (Wireshark, tcpdump, etc.) and data analysis libraries/frameworks in a suitable programming language (Python, Java, C++, etc.).	1	2
	2. Network Capture: Set up a traffic capture interface (e.g. network tap, promiscuous mode) to capture network packets.	3	4
	3. Packet Parsing: Implement code to parse captured packets and extract information like protocol headers, source/destination addresses, ports, and data payloads.	3	4
	4. Data Analysis and Visualization: Use libraries or create visualizations to display network statistics and patterns in an easy-to-understand format.	4	6
	5. Security Monitoring: Develop algorithms or rules to detect suspicious traffic patterns based on common attack methods.	3	5
	6. Documentation: Document the network traffic analyzing activities and related information.	1	2
	7. Presentation: Prepare presentation on the project activities and disseminate during the final exam to the internal and external examiner / assessor.	1	2
	Guiding Principles: • Network neutrality: Respect network traffic without compromising privacy. • Ethical considerations: Obtain necessary permissions before capturing network traffic. • Network performance impact: Minimize the tool's impact on network performance. • Data security: Securely store and analyze captured traffic data.		

Sl.	Experiment name with procedure	Class (3 period)	Marks
4	Project Title: Design and develop an Automated Plant Watering System / Similar any project designed / approved by the guide teacher. Project Requirements: • Hardware: Arduino, Netduino / Raspberry Pi or similar microcontroller, water pump, relay module • Sensors (soil moisture sensor) • Communication protocols (Wi-Fi, Bluetooth, Zigbee) in needed • Software: Arduino IDE, Netduino IDE / Raspberry Pi IDE	16	25
	1. Component Assembly: Connect the soil moisture sensor and water pump to the microcontroller via the relay module.	2	3
	2. Coding: Write a program to read the moisture level and activate the pump when the soil is dry	4	6
	3. Testing: Test the system to ensure the pump activates correctly based on soil moisture.	3	4
	4. Calibration: Adjust the moisture threshold as necessary to optimize watering	3	4
	5. Packaging: Secure all components in a suitable enclosure for outdoor use.	1	3
	6. Deployment: Install the system in a garden and monitor its performance over time.	2	3

	7. Presentation: Prepare presentation on the project activities and disseminate during the final exam to the internal and external examiner / assessor.	1	2
	Guiding Principles: • Device compatibility: Ensure chosen sensors and devices can work seamlessly with the system. • Power efficiency: Optimize energy consumption for battery-powered sensors. • Data security: Encrypt sensitive data transmission if needed. • Scalability: Design the system to accommodate future expansion with additional sensors/devices.		

Sl.	Experiment name with procedure	Class (3 period)	Marks
5	Project Title: Design and develop a Smart Home System / Similar any project designed / approved by the guide teacher. Project Requirements: • Hardware: Arduino / Raspberry Pi or similar microcontroller • Home appliances (lights, thermostats, doors, etc.) • Sensors (temperature, humidity, motion) • Mobile or web platform • Communication protocols (Wi-Fi, Bluetooth, Zigbee)	16	25
	1. Sensor and Device Selection: Choose compatible sensors/devices and a microcontroller board (Arduino, Netduino, ESP32, etc.) based on project scope and affordability	1	2
	2. Component Assembly/ Sensor Interfacing: Learn how to connect sensors to the microcontroller, connect the sensors (temperature, humidity, motion) and home appliances (lights, thermostats, doors, etc.) to the microcontroller, configure sensor readings, and calibrate if needed.	3	4
	3. Device Control: Design code to control relays or other actuators for device operation.	3	4
	4. Communication Protocol Integration: Implement chosen communication protocol library to enable data exchange between sensors, microcontroller, and the user interface.	2	4
	5. Mobile/Web UI Development: Build a user-friendly interface to display sensor readings, control devices, and visualize data (e.g. temperature graphs).	4	5
	6. Packaging and deployment: Secure all components in a suitable enclosure for use and Install the system in a garden and monitor its performance over time.	2	4
	7. Presentation: Prepare presentation on the project activities and disseminate during the final exam to the internal and external examiner / assessor.	1	2
	Guiding Principles: • Device compatibility: Ensure chosen sensors and devices can work seamlessly with the system. • Power efficiency: Optimize energy consumption for battery-powered sensors. • Data security: Encrypt sensitive data transmission if needed. • Scalability: Design the system to accommodate future expansion with additional sensors/devices.		

Sl.	Experiment name with procedure	Class (3 period)	Marks
6.	Project Title: Design and develop a License Plate Recognition (LPR) System/ Similar any project designed / approved by the guide teacher. Project Requirements: ☐ Raspberry Pi with Camera Module ☐ LPR software (Open ALPR is a free option) ☐ Internet Connection (Optional) ☐ Enclosure (Optional)	16	25
	1. Set up Raspberry Pi: Install operating system and LPR software.	2	4
	2. Connect Camera: Configure camera module for proper image capture.	3	4
	3. Train LPR Software: Train software to recognize license plates from your region	4	6
	4. Test and Refine: Test functionality with sample images/videos. Adjust settings for accurate recognition.	2	4
	5. Alerts & Logging: Configure software to send alerts (email/phone) for recognized plates. Implement logging for captured data.	2	3
	6. Enclosure: Build a weatherproof enclosure for outdoor deployment.	2	2
	7. Presentation: Prepare presentation on the project activities and disseminate during the final exam to the internal and external examiner / assessor.	1	2
	Guiding Principles: • Focus: Recognize license plates accurately. • Test & Calibrate: Thoroughly test with diverse images/lighting conditions. Fine-tune software parameters. • Security: Consider data security measures for captured license plates.		

Sl.	Experiment name with procedure	Class (3 period)	Marks
7.	Project Title: Design and develop an Online Course Management System / Similar any project designed / approved by the guide teacher.	16	25
	Project Requirements: • Programming language (e.g. HTML, CSS, JavaScript) • Web server (e.g. Apache) • Database (e.g. MySQL)		
	Steps of Activities: • Design user interface for managing courses, materials, and student enrollment.	2	4
	• Develop backend functionality for storing course data, user accounts, and handling interactions.	4	6
	• Deploy the website to a web server and connect to the database.	4	6
	• Implement features like online quizzes, video lectures, and discussion forums.	2	3
	• Document your work: Maintain clear documentation of your project design, code, and testing procedures.	2	3

	• Presentation: Prepare presentation on the project activities and disseminate during the final exam to the internal and external examiner / assessor.	2	3
	Guiding Principles: • Focus on clarity and functionality: Keep your project focused on achieving a specific goal effectively. • Start small and iterate: Build a basic version first and gradually add features based on testing and feedback. • Consider user experience: Design a user-friendly interface that is easy to navigate and interact with.		

Sl.	Experiment name with procedure	Class (3 period)	Marks
8.	Project Title: Create a basic e-commerce application using Java / Similar any project designed / approved by the guide teacher. Project Requirements: • Frontend: HTML, CSS, JavaScript, JSP (Java Server Pages) • Backend: Java (Servlets, JSP) • Database: MySQL or PostgreSQL • Build Tool: Maven or Gradle • Web Server: Apache Tomcat • IDE: IntelliJ IDEA, Eclipse, or NetBeans	16	25
	1. Prepare Project Structure	2	3
	2. Determine Functional Requirements: User Authentication, Product Management, Shopping Cart, Order Processing	2	3
	3. Determine Detailed Specifications as	4	6
	3.1 User Authentication Registration: Allow users to register by providing username, password, email, and other required details. Login: Authenticate users using the provided username and password. Password Encryption: Use a secure hashing algorithm like BCrypt to encrypt passwords before storing them in the database.		
	3.2 Product Management Product List Page (index.jsp): ▪ Fetch and display all products from the database. ▪ Each product should have a name, price, and a link to the product details page. Product Details Page (product.jsp): ▪ Display detailed information about the selected product. ▪ Provide an option to add the product to the cart.		
	3.3 Shopping Cart • Cart Page (cart.jsp): ▪ Display the list of products added to the cart. ▪ Allow users to update the quantity or remove products from the cart. ▪ Calculate and display the total price.		
	3.4 Order Processing • Checkout Page (checkout.jsp): ▪ Collect user information such as name, address, and payment details. ▪ Validate the information and process the order. ▪ Display an order confirmation message with order details.		
	4. Write Database Schema	3	4

	5. Write data Sample Code Snippets	3	5
	6. Documentation: Prepare the user manual and site development InfoBase.	1	2
	7. Presentation: Prepare presentation on the project activities and disseminate during the final exam to the internal and external examiner / assessor.	1	2
	Guiding Principles: - Secure user authentication: Use strong password hashing techniques - Transaction management: Maintain data consistency during concurrent access. - Data validation: Ensure input accuracy to prevent invalid data. - Backups and recovery: Regularly back up the database for further use and disaster recovery.		

Necessary Resources (Tools, equipment's and Machinery):

- As per individual Project requirement

Recommended Books:

Sl	Book Name	Writer Name	Publisher Name & Edition
01	Successful ICT Projects in Access	P. M Heathcote	Payne-Gallway Publishers
02	Oracle PL/SQL Programming	Steven Feuerstein, Bill Pribyl, Chip Dawes	O'Reilly Media
03	Data Communications and Networking	Behrouz A. Forouza	McGraw Hill 5th Edition
04	Introduction to Sensors in IoT and Cloud Computing Applications	Ambika Nagaraj	Bentham Books
05	PIC Microcontroller and Embedded system	Muhammad Ali Mazid	Pearson
06	Digital Video Surveillance and Security	Anthony C. Caputo	Elseiver
07	Beginning HTML, XHTML, CSS, and JavaScript	Jon Duckett	Wrox 1st Edition

Website References:

Sl	Web Link	Remarks
01	What is a DBMS (Database Management System)? Splunk	
02	Website: https://en.wikipedia.org/wiki/Computer_network	
03	https://copyprogramming.com/howto/internet-of-things-a-hands-on-approach-pdf	
04	www.microchip.com	
05	https://www.dahuasecurity.com/support/training/eLearning	
06	https://figma.com/	